

Project Coversheet

Full Name	Purvi Rajes Tulcidas
Project Title (Example – Week1, Week2, Week3, Week 4)	Week4

Instructions:

Students must download this cover sheet, use it as the first page of their project, and then save the entire document as a PDF before submission.

Project Guidelines and Rules

1. Formatting and Submission

- Format: Use a readable font (e.g., Arial/Times New Roman), size 12, 1.5 line spacing.
- Title: Include Week and Title (Example - Week 1: Travel Ease Case Study.)
- File Format: Submit as PDF or Word file
- Page Limit: 4–5 pages, including the title and references.

2. Answer Requirements

- Word Count: Each answer should be within 100–150 words; Maximum 800–1,200 words.
- Clarity: Write concise, structured answers with key points.
- Tone: Use formal, professional language.

3. Content Rules

- Answer all questions thoroughly, referencing case study concepts.

- Use examples where possible (e.g., risk assessment techniques).
- Break complex answers into bullet points or lists.

4. Plagiarism Policy

- Submit original work; no copy-pasting.
- Cite external material in a consistent format (e.g., APA, MLA).

5. Evaluation Criteria

- Understanding: Clear grasp of business analysis principles.
- Application: Effective use of concepts like cost-benefit analysis and Agile/Waterfall.
- Clarity: Logical, well-structured responses.
- Creativity: Innovative problem-solving and examples.
- Completeness: Answer all questions within the word limit.

6. Deadlines and Late Submissions

- Deadline: Submit on time; trainees who fail to submit the project will miss the “Certificate of Excellence”

7. Additional Resources

- Refer to lecture notes and recommended readings.
- Contact the instructor or peers for clarifications before the deadline.

Week 4: Project - Business Intelligence Dashboard for TechHub Retail

Tableau Public URL: [TechHub Tableau Dashboard | Tableau Public](#)

Has sheets on top to navigate through various worksheets and 2 dashboards, 1 for KPIs and the other for business performance

Introduction

TechHub Retail is a rapidly growing UK-based online electronics retailer that's has experienced significant expansion over the past 18 months. Despite this growth, the organisation lacks a consolidated analytics capability to support executive-level strategic planning for 2025. As a result, senior leadership has limited visibility into performance drivers across products, customers, suppliers and regions.

The purpose if this project is to design an interactive executive dashboard that transforms operational data into actionable business insights. The analysis focuses on identifying growth opportunities, understanding performance trends and seasonality, evaluating customer and product profitability, and supporting forward-looking strategic decision-making.

By integrating sales, customer and product datasets within Tableau Public, this report provides data-driven insights and strategic recommendations that enable the Chief Executive and leadership team to make informed decisions aligned with TechHub Retail's 2025 growth objectives.

Multi-Dataset Integration Summary

This analysis integrates three core datasets representing TechHub Retail's business operations:

- Sales Data – transactional information including revenue, quantity, cost price and order dates

- Customer Data – demographic and behavioural attributes such as age group, loyalty tier, signup date and location
- Product Data – product category, supplier and launch date information

The datasets were connected within Tableau using logical relationships, linking Sales to Customers via *Customer ID* and Sales to Products via *Product ID*. This approach preserves datasets granularity while allowing flexible aggregation across dimensions, avoiding duplication and aggregation errors with physical joins.

Several calculated fields were created to support business analysis, including Profit Amount, Profit Margin (%), Average Order Value, Customer Lifetime Value, Revenue per Customer, New Customer and Product Age (Days). These metrics enable profitability, lifecycle and segmentation analysis not available in the raw data.

Minor challenges included missing geographic values and varying data granularity, which were addressed through filtering, aggregation and transparent reporting of limitations.

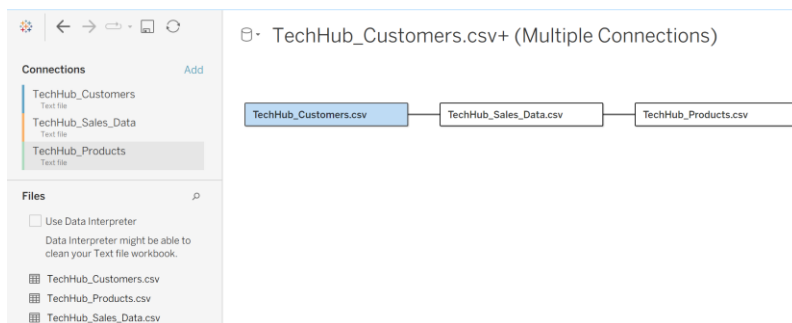


Figure 1: Tableau Data Source Relationship view

Dashboard Design Summary

Two dashboards were developed to balance executive clarity with analytical depth.

Dashboard 1 – Executive KPI Dashboard

this dashboard provides a high-level snapshot of overall business performance. Six KPIs are displayed prominently:

- Purpose: provide a high-level snapshot of overall business performance

- Audience: chief executive and senior leadership
- Focus: immediate performance indicators

Key KPIs included:

- Total Revenue
- Average Order Value
- Total Customers
- Customer Acquisition Rate
- Average Profit Margin
- Top Customer Segment

The design enables executives to assess business health within seconds without requiring technical interpretation.

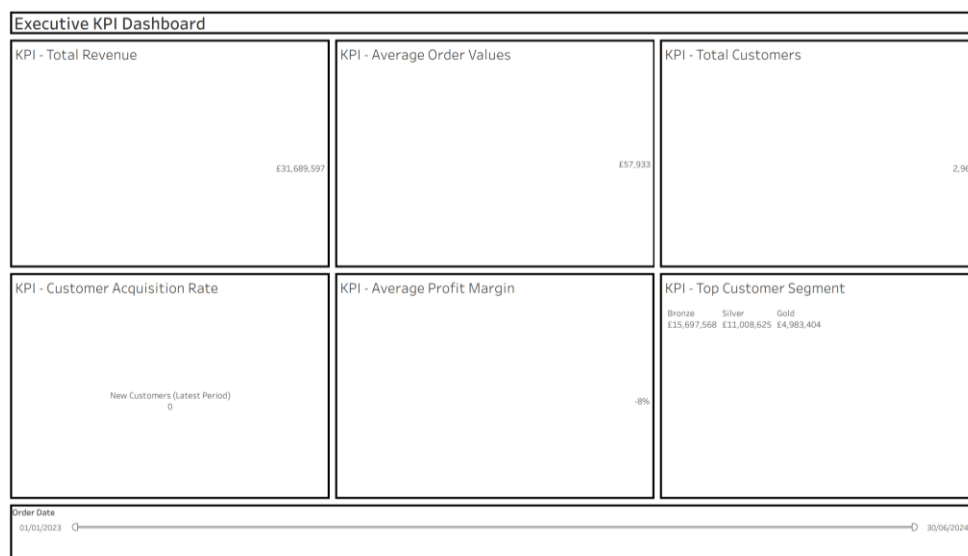


Figure 2: Executive KPI Dashboard

Dashboard 2 – Business Performance Dashboard

This dashboard supports deeper analysis through the following visualisations:

- Purpose: enable exploratory analysis and strategic insight
- Audience: executives and senior analysts
- Focus: trends, segmentation, geography and performance drivers

Key visualisations included:

- Sales & Profitability Trends (revenue vs profit over time)
- Geographic Performance (revenue per customer by region and city)
- Product Portfolio Analysis (treemap sized by revenue and coloured by profit margin)
- Customer Segmentation Matrix (lifetime value vs tenure)
- Supplier & Product Performance (supplier revenue and product count)

Interactive filters for date range, product category, loyalty tier, and geography allow dynamic exploration while maintaining visual clarity

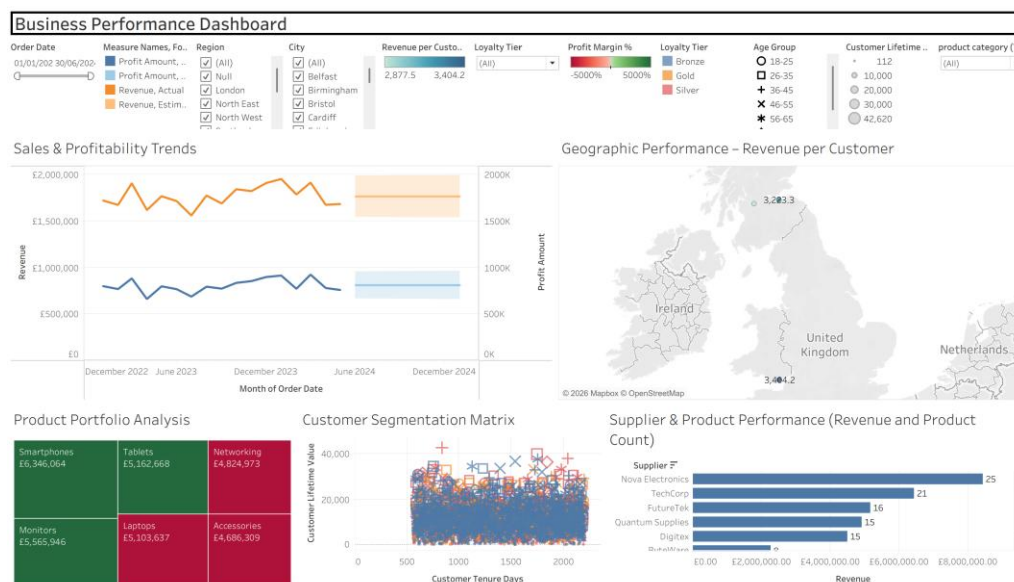


Figure 3: Business Performance Dashboard

Dashboard Design Rationale:

The dashboard layout follows best-practice executive design principles:

- Filters placed at the top to avoid visual clutter
- Minimal colour palette focused on performance interpretation
- KPI in one dashboard and visuals in a different dashboard

This structure enables rapid insight generation while preserving analytical depth for strategic planning.

Key Insights & Findings

Analysis reveals that revenue performance is stable, but profitability varies significantly across products, suppliers, and customer segments. Profit trends fluctuate more than revenue, indicating margin pressure driven by product mix and cost structure.

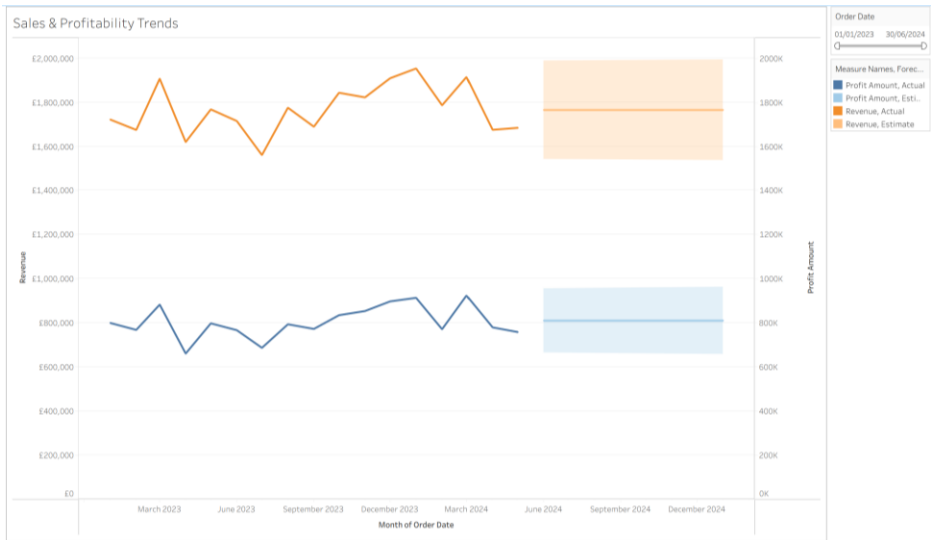


Figure 4: Sales & Profitability Trends chart

Product analysis shows that Smartphones, Tablets, and Monitors combine strong revenue contribution with healthier profit margins, while categories such as Laptops and Accessories generate volume but underperform on margin. Newer products consistently outperform older ones, highlighting the importance of lifecycle management.



Figure 5: Product Portfolio Analysis treemap

Customer analysis demonstrates a clear relationship between tenure, loyalty tier, and lifetime value. While Bronze and Silver customers contribute the highest total revenue, Gold-tier customers generate significantly higher value per customer. Geographic analysis indicates that urban regions deliver stronger revenue per customer than rural areas.

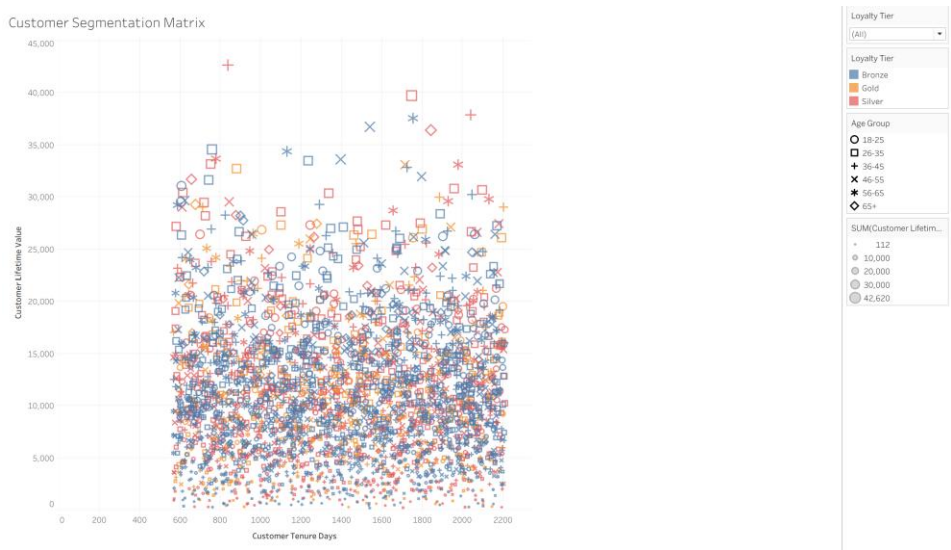


Figure 6: Customer Segmentation Matrix and Loyalty Tier KPI



Figure 7: Geographic Performance - Revenue per Customer map

Supplier performance is concentrated among a small number of high-efficiency suppliers, revealing opportunities to rationalise underperforming supplier relationships.

Overall, the findings suggest that TechHub’s growth opportunity lies not in increasing volume, but in optimising margins, retention, and product strategy.

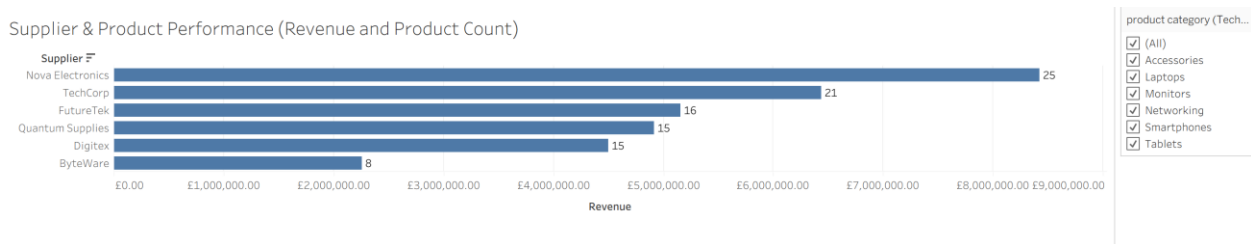


Figure 8: Supplier & Product Performance bar chart

Business Questions

1. Which product categories and suppliers offer the best profit margins for 2025 focus?

Smartphones, Tablets, and Monitors consistently deliver higher profit margins alongside strong revenue performance. Supplier analysis shows that a small group of suppliers generate disproportionate revenue with fewer products, indicating superior efficiency. These categories and suppliers represent the strongest candidates for strategic focus.

2. How do customer demographics (age, location, loyalty tier) impact purchasing behaviour?

Customer behaviour varies notably by loyalty tier, age, and location. Gold-tier customers generate higher lifetime value, while younger age groups drive transaction volume. Urban customers exhibit higher revenue per customer, suggesting stronger purchasing power and upsell potential.

3. What seasonal patterns exist across different product categories and regions?

Moderate seasonality is observed, with higher revenue during late Q3 and Q4. Core technology categories display relatively consistent demand throughout the year, supporting stable forecasting assumptions.

4. Which customer acquisition channels deliver the highest lifetime value customers?

While acquisition channel data is unavailable, customers with longer tenure and higher loyalty tiers demonstrate significantly higher lifetime value. This suggests that

acquisition strategies attracting high-intent customers outperform volume-driven approaches.

5. How does product age (time since launch) correlate with sales performance?

Product age shows a negative correlation with sales and profitability. Newer products outperform older SKUs, particularly in fast-moving technology categories.

6. What are the top 3 strategic recommendations based on integrated data analysis?

Margin optimisation, customer retention and loyalty tier progression, and supplier efficiency emerge as the most impactful strategic priorities for 2025.

Strategic Recommendations

Recommendation 1: Prioritise High-Margin Product Categories

Focus investment on categories demonstrating both revenue scale and strong margins while reviewing pricing or discontinuation strategies for low-margin products.

- Impact: Improved profitability and inventory efficiency
- Timeline: 3–6 months

Recommendation 2: Strengthen Customer Retention and Loyalty Progression

Shift marketing focus toward retaining existing customers and upgrading loyalty tiers rather than maximising acquisition volume.

- Impact: Higher lifetime value and reduced acquisition costs
- Timeline: 3–9 months

Recommendation 3: Optimise Supplier Portfolio

Consolidate supplier relationships around high-efficiency partners and reduce dependency on low-performing suppliers.

- Impact: Improved supply chain efficiency and margin control
- Timeline: 6–12 months

These actions directly support TechHub Retail's objective of sustainable, data-driven growth in 2025.

Critical Reflection

The dashboards effectively support executive decision-making by combining high-level KPIs with interactive analytical views. Executives can rapidly assess performance while retaining the ability to explore drivers and scenarios.

Future BI initiatives could benefit from incorporating predictive analytics, improved geographic data quality, and marketing channel information to enhance attribution analysis. Simplifying visible controls further could also reduce cognitive load for non-technical users.

Overall, the dashboard provides a strong foundation for strategic planning and demonstrates the value of integrated BI at the executive level.

Data Issues or Risks

Several data limitations should be noted:

- Missing or unknown geographic values limit full regional accuracy
- Absence of acquisition channel data restricts direct attribution analysis
- Aggregated metrics may obscure niche or outlier behaviour
- External market factors are not explicitly captured

Despite these limitations, the data is sufficient for executive-level strategic analysis, and identified risks do not materially undermine the conclusions.